Homework 6

October 23, 2026

Name: YuanHao Tang Student ID: 3120265621

Problem 1

Let $\mathcal{A}$ be a linear transformation from a four dimensional linear space $\mathcal{V}$ to itself, $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ be a basis of the linear space $\mathcal{V}$, and the matrix representation of $\mathcal{A}$ with respect to this basis be

$$A = \begin{pmatrix} 2 & 0 & 2 & 1 \\ 6 & 1 & 4 & 4 \\ 10 & 0 & 0 & 4 \\ 7 & 0 & -7 & 2 \end{pmatrix}.$$

Find out

(1) the eigenvalues and eigenvectors of matrix A;

(2) the eigenvalues and associated eigenvectors of $\mathcal{A}$.

Solution

(1) The characteristic polynomial is $\det(\lambda I - A) = (\lambda - 1)^3(\lambda - 2)$. Both $A - I$ and $A - 2I$ have rank three. All eigenvectors for eigenvalues $1, 2$, respectively, are

$$\boxed{x_1 = c(0, 1, 0, 0)^T, \qquad x_2 = c(1, 2, 1, -2)^T, \qquad c \neq 0.}$$

(2) The eigenvalues are again $1, 2$. Converting coordinates to vectors in the original space, all associated eigenvectors are

$$\boxed{v_1 = c\alpha_2, \qquad v_2 = c(\alpha_1 + 2\alpha_2 + \alpha_3 - 2\alpha_4), \qquad c \neq 0.}$$

Problem 2

Let

$$A = \begin{pmatrix} 4 & 6 & 0 \\ -3 & -5 & 0 \\ -3 & -6 & 1 \end{pmatrix}.$$

Find out the invariant polynomials of A, the natural normal form N of A, and the Jordan canonical forms of A (the complex case and the real case, respectively).

Solution

For $\lambda I - A$, $D_1 = 1$. Since $\operatorname{rank}(A - I) = 1$, every second-order minor contains $\lambda - 1$, and one is $-6(\lambda - 1)$, so $D_2 = \lambda - 1$. Also $D_3 = \det(\lambda I - A) = (\lambda - 1)^2(\lambda + 2)$. Hence the invariant polynomials are

$$\boxed{d_1 = 1, \qquad d_2 = \lambda - 1, \qquad d_3 = (\lambda - 1)(\lambda + 2)}.$$

Discard the unit entry and form column companion blocks for the nonconstant entries:

$$\boxed{N = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 1 & -1 \end{pmatrix}}.$$

The eigenvalue 1 has geometric multiplicity two, while -2 is a simple root. Thus both Jordan forms are

$$\boxed{J_{\text{complex}} = J_{\text{real}} = \operatorname{diag}(1, 1, -2)}.$$

Problem 3

Let A be a matrix whose elementary divisors are

$$(\lambda+1),\quad (\lambda+1)^2,\quad (\lambda+1+j),\quad (\lambda+1-j),$$
$$(\lambda+1)^2,\quad (\lambda+1+j)^2,\quad (\lambda+1-j)^2.$$

Find out

(1) the Smith canonical form S of the eigenmatrix $\lambda E - A$;

(2) the natural normal form N, the complex Jordan canonical form J_{complex} and the real Jordan canonical form J_{real} of A.

Solution

Put $j^2 = -1$, $x = \lambda + 1$ and $q = x^2 + 1$. The elementary divisors have total degree eleven.

(1) Sort the exponents for each root, padding with zeros on the left to length three:

Root	1	2	3
-1	1	2	2
$-1-j$	0	1	2
$-1+j$	0	1	2

The nonconstant invariant polynomials are $\psi_1 = x$, $\psi_2 = x^2 q$ and $\psi_3 = x^2 q^2$. Thus

$$\boxed{S = \operatorname{diag}(I_8, \psi_1, \psi_2, \psi_3)}.$$

(2) For monic $p = \lambda^m + c_{m-1}\lambda^{m-1} + \cdots + c_0$, let $C(p)$ be its column companion matrix: ones on the subdiagonal, last column $-(c_0, \ldots, c_{m-1})^T$, and zeros elsewhere. Then

$$\boxed{N = \operatorname{diag}\big(-1, C(\psi_2), C(\psi_3)\big)},$$

where

$$\psi_2 = \lambda^4 + 4\lambda^3 + 7\lambda^2 + 6\lambda + 2,$$
$$\psi_3 = \lambda^6 + 6\lambda^5 + 17\lambda^4 + 28\lambda^3 + 28\lambda^2 + 16\lambda + 4.$$

Define

$$J_2(\rho) = \begin{pmatrix} \rho & 1 \\ 0 & \rho \end{pmatrix}, \qquad K_\pm = \operatorname{diag}\big(-1 \pm j, J_2(-1 \pm j)\big).$$

Each elementary divisor gives one Jordan block, so

$$\boxed{J_{\text{complex}} = \operatorname{diag}\big(-1, J_2(-1), J_2(-1), K_-, K_+\big)}.$$

For the real form, pair conjugate blocks. Put

$$R = \begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix}, \qquad R_2 = \begin{pmatrix} R & I_2 \\ 0 & R \end{pmatrix}.$$

Then

$$\boxed{J_{\text{real}} = \operatorname{diag}\big(-1, J_2(-1), J_2(-1), R, R_2\big)}.$$

All these forms are 11×11; Jordan blocks may be reordered.